1. Identity

Document	STAR-PKG-STAR-ALBERS-EN · v2.0 · 2026-10-21
Package	star-albers
Version	0.1.0
Capability	star.geodesy.albers-conic
Maturity	RELEASED
Python	⩾3.9
Source commit	94fcf7c
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_albers

2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_albers-0.1.0-py3-none-any.whl
```

FILE	SHA-256
star_albers-0.1.0-py3-none-any.whl	a428fb35be3fce23b5fcbd3243a9ffaba6a8a828d1344a130d2c2b9a4b04ecb9
star_albers.py	2526228dcff2e0540ecc506cc8f42ccd770a5b0f4261beb2806047b7694011c8
star_albers-0.1.0-source.zip	81d78cfe6f02a071938142abce39fb26e19e0ae61cc67b198b150111421a361e

5. What it does, and its limits

Albers equal-area conic projection, forward, inverse and both scale factors, for cones in either hemisphere and the tangent cone; within 1.9e-6 m of PROJ and 9.1e-7 m of PyGeodesy on 600 cases; reproduces Snyder's published coordinates and scale factors and refuses degenerate cones.

No false easting or northing, no named national grids. The cone constant is computed as a difference between the two standard parallels: when they are very close (0.04 deg apart in the worst case measured) about four digits are lost and the difference from the libraries reaches 2 micrometres; PyGeodesy's formulation does not have this loss. The accuracy stated is agreement with two libraries, not with a survey.

- `albers_forward` and `albers_inverse`: Snyder's formulas (USGS PP 1395, 14-1 ... 14-21), cones in either hemisphere, equal parallels for the tangent cone, the central meridian anywhere including across the date line. Reproduce Snyder's numerical example to the printed 0.1 m.
- `albers_scale` returns (h, k) with h·k = 1: areas are preserved; k = 1 on the standard parallels. Reproduces Snyder's h = 1.0085173 and k = 0.9915546.
- Measured on 600 cones and points on WGS-84 (60 tangent cones): forward within 1.9e-6 m of PROJ and 9.1e-7 m of PyGeodesy; inverse within 2.6e-6 m of the starting point; parallel scale within 9e-14 of PyGeodesy and 1.3e-9 of PROJ (which differentiates numerically).
- Every function returns finite floats or raises `ValueError`: non-numeric, boolean, NaN or infinite input, latitudes beyond ±89, longitudes beyond ±180, standard parallels on opposite sides of the equator or within 1e-3 deg of it, an ellipsoid outside the stated ranges, a point outside the image of the cone.

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_star_albers_0_1_0,
  author  = {Cavazzini, Andrea},
  title   = {{star-albers}: Albers equal-area conic projection},
  version = {0.1.0},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_albers/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

Related: space activities and export control · acceptable use · evidence and limits · licence texts